

Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 23 papers, 2070–2082 BS

8. Wireless Systems and Standards

6 hrs · asked in **22 of 23 papers** (only 72 Ma skips it) · 5 syllabus sub-topics · **283 marks, 13.9% of the archive** · **no numerical component**: the one calculation the syllabus touches, GSM frame efficiency, is worked in **Ch 7 companion §3**.

Read this first:

- This chapter is **always the last question on the paper**, and it is nearly always a full 8 marks plus a 4 or 5 mark rider. Two of its topics are among the six most repeated in the whole subject.
- Five blocks carry the chapter, in order of return:
 - **Regulatory issues** **TOP 9/23** → pure recall, no diagram, no formula. **The cheapest 4 to 5 marks in the subject.**
 - **GSM traffic and control channels** **TOP 9/23** → the BCH / CCCH / DCCH tree, with what each channel does and which way it points.
 - **GSM system architecture** **TOP 7/23** → one diagram plus one line per block.
 - **IS-95 forward and reverse channels** **TOP 6/23** → draw the composite block diagram and name the four forward, two reverse channels.
 - **GSM frame structure and hierarchy** **HOT 5/23** → hyperframe down to time slot, with the four durations.
- **Learn 4 diagrams cold**: GSM architecture, GSM frame hierarchy, the forward CDMA (IS-95) channel, and the GSM signal-processing chain. They answer about 30 marks.
- **Name clash to watch**. In this chapter **BCH** = **broadcast channel** and **SCH** = **synchronisation channel**. In **Ch 6** BCH is the Bose–Chaudhuri–Hocquenghem block code. Same letters, unrelated.
- GSM's access scheme (TDMA/FDMA/FDD with slow hopping) is argued in **Ch 7**; the frame efficiency proof is in **Ch 7 companion §3**. **Do not re-derive either here**, cross-refer and spend the words on the channels.
- **WLAN and WiFi** are answered in **Ch 7 §7.11**. **WiMAX, LTE, UMTS and EVDO belong here**, in §8.13.

8.1 GSM: what it is, its services and its features

What is GSM? State its services and features. Give its specifications **5**

74 Bh **5** (specifications of GSM) · never asked alone as a definition, but **every GSM question opens with these three lines**, so they are worth two minutes

Definition and history

- **GSM** = global system for mobile communications.
- **The first cellular system to specify digital modulation, network-level architecture and services together**, not just an air interface.
- **1982**: CEPT sets up *Groupe Spécial Mobile* to build one pan-European cellular system in the 900 MHz band.
- **1990**: the same specification is applied at 1800 MHz as **DCS 1800**, a PCN deployment started in the UK.
- **1992**: renamed **Global System for Mobile Communications**, for marketing reasons. The acronym stayed.
- Later carried into the USA at 1900 MHz as **PCS 1900**.

Features worth naming **2**

- **Superior speech quality** against the analog systems it replaced.
- **High security**: SIM-based authentication and over-the-air ciphering.
- **International roaming**, because one standard covers many countries.
- **Low-power hand-held terminals**, helped by discontinuous transmission.
- **Open, published standard** → many vendors, low equipment cost.

Three categories of service

1. **Teleservices:** voice, emergency calling, **SMS**, fax.
2. **Bearer services:** data transport, **300 to 9600 bps**.
3. **Supplementary services:** call forwarding, call barring, call waiting, caller ID, conferencing.

The three frequency plans

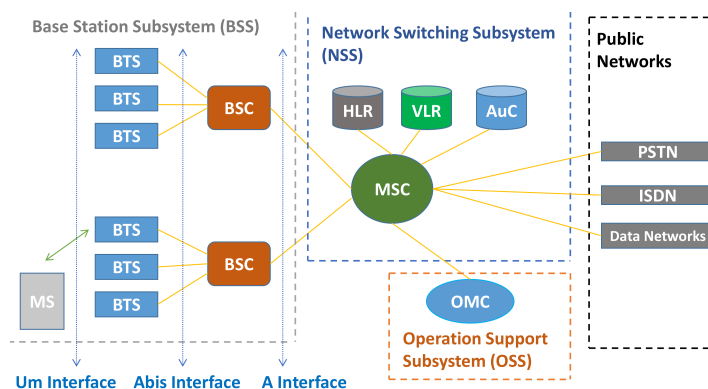
System	Uplink (MS → BS)	Downlink (BS → MS)
GSM 900	890–915 MHz	935–960 MHz
DCS 1800	1710–1785 MHz	1805–1880 MHz
PCS 1900	1850–1910 MHz	1930–1990 MHz

All three are FDD. **Uplink is always the lower band**, because path loss is slightly lower there and the handset is the power-limited end.
SST Ch 8

8.2 GSM system architecture one of the two most repeated questions in this chapter

Draw and explain the GSM architecture; explain the operation of each component 4 5 7 8 4+4

TOP 7/23 82 Bh 4 (NSS only), 80 Ba 8, 79 Bh 8, 73 Bh 4 (NSS) + 5, 73 Ma 7, 71 Ma 4+4, 70 Ma 8 · four papers ask only for the NSS, so learn that box in more detail than the rest



Draw exactly this. Three dashed boxes, three named interfaces, and the public networks hanging off the MSC. SST Ch 8

1. Base Station Subsystem (BSS), the radio subsystem

- **MS** = mobile station = **ME** (mobile equipment) + **SIM**.
 - ME carries the **IMEI**, the equipment's own identity.
 - SIM carries the **IMSI**, the subscriber's identity, plus the secret key.
 - **Identity is in the card, not the handset** → personal mobility.
- **BTS** = base transceiver station. Houses the radio transceivers that define one cell. Talks to the MS over the **Um** air interface.
- **BSC** = base station controller.
 - Manages radio resources for one to several hundred BTSs.
 - Radio channel setup, frequency hopping, power control.
 - **Performs handover between two BTSs under itself**; anything wider goes to the MSC.
 - Connects to the BTS over the **Abis** interface.

3. Operation Support Subsystem (OSS) 2

- Built around one or more **OMC** = operation and maintenance centres.
- Lets engineers **monitor, diagnose and troubleshoot** every part of the network: MS, BTS, BSC, MSC.
- Maintains hardware and network performance records.

2. Network and Switching Subsystem (NSS) 4

- Handles **switching of GSM calls between the BSCs and external networks**, and holds every subscriber database.
- **MSC** = mobile switching centre. The central unit: controls traffic among BSCs, sets up and releases calls, handles inter-BSC handover, and gateways to **PSTN, ISDN and data networks**. Reaches the BSC over the **A** interface.
- **HLR** = home location register.
 - Permanent record of every subscriber **homed** on this MSC: **IMSI**, number, subscribed services.
 - Also holds the subscriber's **current location**, so an incoming call can be routed to wherever the mobile now is.
- **VLR** = visitor location register.
 - Temporarily stores the **IMSI** and customer data of every **roaming** subscriber inside this MSC's area.
 - On log-in the MSC tells the visitor's **home HLR** where they are, so the home HLR can route incoming calls over the PSTN.
- **AuC** = authentication centre. Strongly protected database of the authentication and ciphering keys of every subscriber in the HLR and VLR.
- **EIR** = equipment identity register.
 - Lists **IMEIs** as white (allowed), grey (tracked) or black (barred).
 - A stolen handset's **IMEI** (dial **##06#** to read it) is blacklisted here, and optionally in the **CEIR**, which bars it on every operator that subscribes.
 - **EIR bars the handset, AuC authenticates the subscriber**. Different targets.

- Handles **charging and billing**.

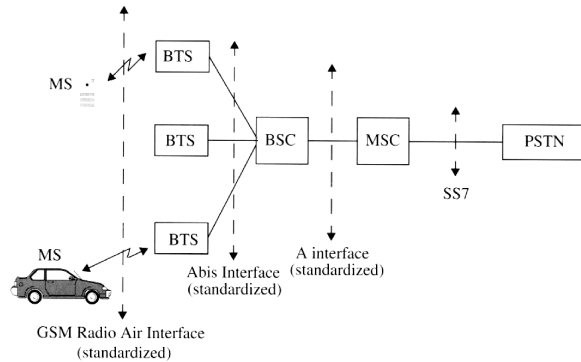


Figure 11.6 The various interfaces used in GSM.

The three standardised interfaces. **Naming them is usually worth a mark on its own**, and they are what makes multi-vendor GSM possible. SST Ch 8, Fig 11.6

The interfaces, and why they matter

- **Um** = MS to BTS, the **air interface**. The only radio hop.
- **Abis** = BTS to BSC. Standardised, so a BTS and a BSC may come from different vendors.
- **A** = BSC to MSC. Standardised.
- **SS7** carries the signalling on from the MSC into the PSTN.
- **Because Abis and A are open, an operator can mix vendors**. That is the whole commercial point of GSM being a *network standard* and not only an air interface.

Answer skeleton 8

1. Draw the figure. Three subsystems boxed, interfaces labelled. 3
2. BSS: MS, BTS, BSC, one line each. 2
3. NSS: MSC, HLR, VLR, AuC, EIR, one line each. 2
4. OSS: OMC, monitoring and billing. 1

8.3 GSM radio subsystem and specifications

Specifications of GSM; the radio subsystem and ARFCN 5

74 Bh 5 · **these numbers also feed the frame-structure and frame-efficiency answers**, so they earn their place even though the question itself is rare

The radio plan

- Two **25 MHz** bands: **890–915 MHz reverse** (MS to BS), **935–960 MHz forward** (BS to MS).
- Multiple access is **FDD + TDMA over FDMA**, with optional **slow frequency hopping**. See Ch 7 §7.3.
- Each band is cut into **200 kHz** carriers called **ARFCNs**, absolute radio frequency channel numbers.
- **One ARFCN names a forward and reverse pair 45 MHz apart**, so an ARFCN is a duplex channel, not a single frequency.
- 25 MHz/200 kHz = 125 carriers, but a **100 kHz guard band** is left at each edge, so only **124 duplex channels** are usable, ARFCN 1 to 124.
- Each 200 kHz carrier is time shared by **8 users** using TDMA.
- **Physical channel = (ARFCN, time slot number)**. A **logical channel** is then mapped onto a physical channel, see §8.5.
- DCS 1800 uses 1710–1785 / 1805–1880 MHz, three times the spectrum, giving **374 duplex channels**.

Table 10.3 GSM Air Interface Specifications Summary

Parameter	Specifications
Reverse Channel Frequency	890 - 915 MHz
Forward Channel Frequency	935 - 960 MHz
ARFCN Number	0 to 124 and 975 to 1023
Tx/Rx Frequency Spacing	45 MHz
Tx/Rx Time Slot Spacing	3 Time slots
Modulation Data Rate	270.833333 kbps
Frame Period	4.615 ms
Users per Frame (Full Rate)	8
Time slot Period	576.9 μs
Bit Period	3.692 μs
Modulation	0.3 GMSK
ARFCN Channel Spacing	200 kHz
Interleaving (max. delay)	40 ms
Voice Coder Bit Rate	13.4 kbps

Reproduce this table for a 5 specifications question. Rappaport Table 10.3, via SST Ch 8 · **one printed value to know about**: the table gives the voice coder at 13.4 kbps, while the RPE-LTP codec itself produces 260 bits per 20 ms = 13.0 kbps exactly (§8.7). Quote 13 kbps if you are asked for the codec rate. **verified/added**

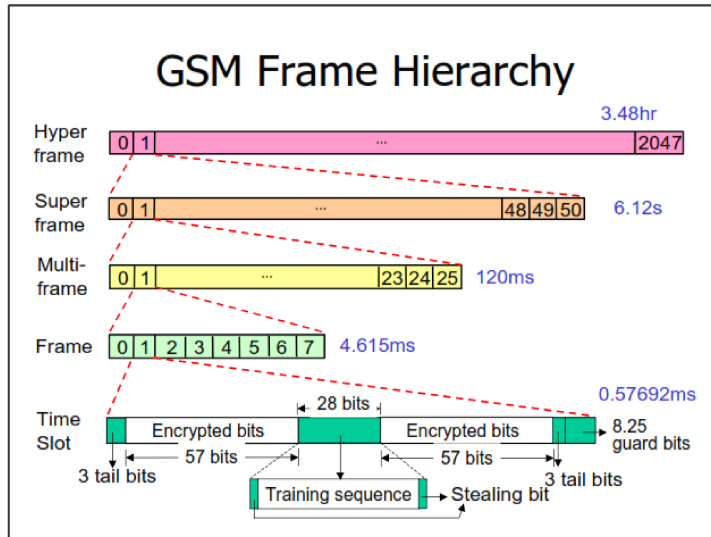
Rates worth memorising

- Modulation **0.3 GMSK**, channel data rate **270.833 kbps** per carrier.
- Bandwidth efficiency = $270.833/200 = 1.354$ bps/Hz.
- Per-user channel rate = $270.833/8 = 33.854$ kbps.
- After all overhead, **maximum user data rate 24.7 kbps**.
- Bit period **3.692 μs**, time slot **576.9 μs**, frame **4.615 ms**.

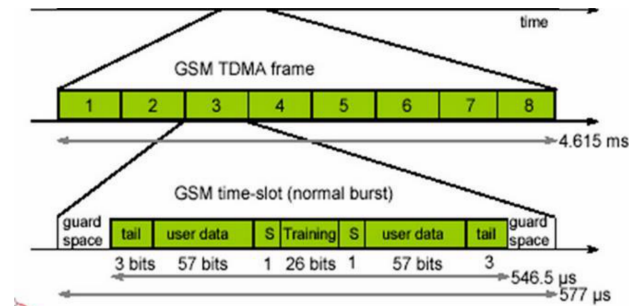
8.4 GSM frame structure and frame hierarchy

Draw and explain the GSM frame structure / channel structure / frame hierarchy 4 5 6

HOT 5/23 82 Bh 4 (frame hierarchy), 77 Ch 4, 76 Bh 5, 74 Ma 6 (channel structure), 70 Bh 4 · 74 Ma adds “show that TDMA frame efficiency cannot reach 100 % in GSM” 2, worked in full in Ch 7 companion §3



Draw the hierarchy top down and label the four durations. That is the whole answer to a 4 or 5 question. SST Ch 8



One frame down to one normal burst. The 546.5 μs is the useful part, 577 μs is the whole slot including guard space. SST Ch 8

The five levels, bottom up

1. Time slot = 156.25 bits = 576.92 μs. One burst.
2. TDMA frame = 8 slots = 1250 bits = 4.615 ms (= 120/26 ms).
3. Multiframe, two kinds:
 - 26-frame traffic multiframe = 120 ms. Carries TCH, SACCH, FACCH.
 - 51-frame control multiframe = 235.4 ms. Carries BCH and CCCH.
4. Superframe = 51 traffic multiframe = 26 control multiframe = 1326 TDMA frames = 6.12 s.
5. Hyperframe = 2048 superframes = 2 715 648 TDMA frames = 3 h 28 min 53.76 s.

Why a hyperframe at all: the ciphering algorithm and the hopping sequence are both driven by the frame number, so the frame number must not repeat for a long time. · Two multiframe sizes, one superframe: 51 × 26 either way = 1326, which is exactly why 26 and 51 were chosen. verified/added · The SST slide prints the hyperframe as 3 h 28 min 52.76 s; 2048 × 6.12 = 12533.76 s recomputes to 53.76 s. verified/added

Inside a normal burst 3

- 3 tail bits (0, 0, 0) → mark the start, let the equalizer settle.
- 57 encrypted data bits.
- 1 stealing flag → says whether this half is user data or FACCH signalling.
- 26 training bits → a known sequence the receiver correlates against to build a channel model for the equalizer (Ch 5).
- 1 stealing flag, 57 encrypted data bits, 3 tail bits.
- 8.25 guard bits → stop bursts from overlapping and give the power amplifier ramp-up and ramp-down time.
- Total 3 + 57 + 1 + 26 + 1 + 57 + 3 = 148 useful + 8.25 guard = 156.25 bits.

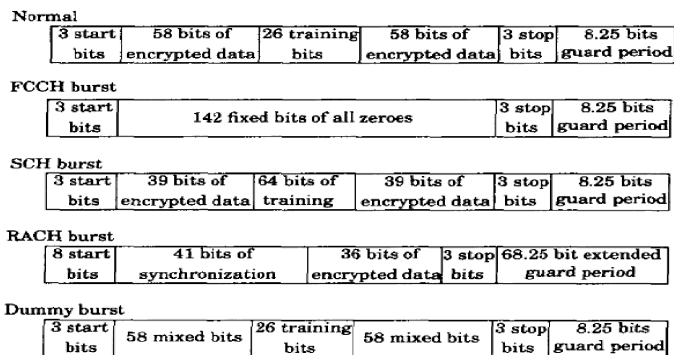


Figure 10.9
Time slot data bursts in GSM.

Five burst formats, all 156.25 bits. Rappaport Fig 10.9, via SST Ch 8

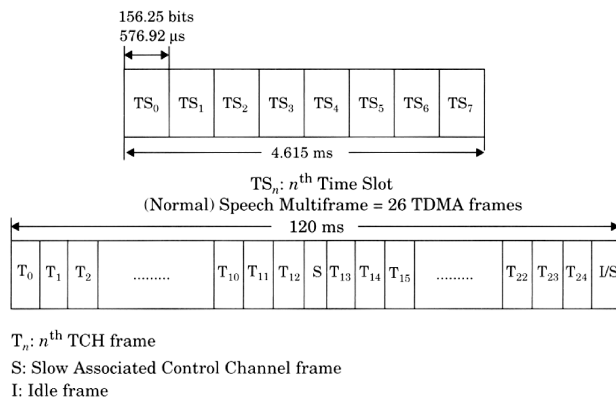


Figure 11.7 The speech dedicated control channel frame and multiframe structure.

Traffic (speech) multiframe, 26 frames, 120 ms. SST Ch 8, Fig 11.7

Traffic multiframe 2

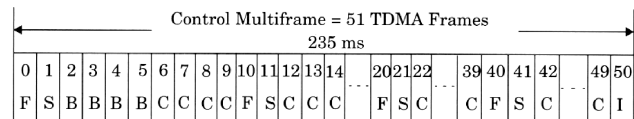
- TCH data runs in **consecutive frames**, but **frames 13 and 26 are not TCH**.
- **Frame 13** carries **SACCH**.
- **Frame 26** carries **idle** bits for a full-rate TCH, and SACCH for a half-rate TCH.
- The idle frame is what gives the mobile time to **measure neighbouring cells** for MAHO (Ch 2).

Frame efficiency, in one line 2

- $\eta_f = \left(1 - \frac{b_{OH}}{b_T}\right) \times 100\%$, with $b_T = 8 \times 156.25 = 1250$ bits and $b_{OH} = 8(6 + 8.25 + 26) = 322$ bits $\rightarrow \eta_f = 74.24\%$.
- **It can never be 100 %**: tail, training and guard bits are needed for the equalizer and for timing, and removing them breaks the link.
- Full working, with the 74 Ma and 70 Bh statements, is in Ch 7 companion §3. Do not redo it here.

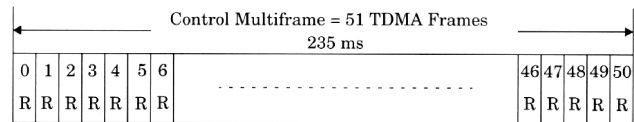
Why five formats

- **Normal burst** $\rightarrow$ TCH and most control channels.
- **FCCH burst** $\rightarrow$ 142 fixed bits, **all zeros**. Under GMSK an all-zero block becomes a pure tone, which is exactly what a mobile needs to pull its local oscillator onto the base station.
- **SCH burst** $\rightarrow$ a long **64-bit training sequence**, because the mobile has no channel estimate yet.
- **RACH (access) burst** $\rightarrow$ 8 start bits and a **68.25-bit extended guard period**. The mobile does not yet know its timing advance, so the burst is kept short to guarantee it cannot land on the next slot.
- **Dummy burst** $\rightarrow$ mixed bits, no information. Sent on unused slots of the BCH carrier so its **power stays constant** and neighbouring mobiles can measure it.



F : FCCH burst (BCH)
S : SCH burst (BCH)
B : BCCH burst (BCH)
C : PCH/AGCH burst (CCCH)
I : Idle

(a)



R : Reverse RACH burst (CCCH)

(b)

Figure 11.8 (a) The Control Channel Multiframe (forward link for TS0); (b) The Control Channel Multiframe (reverse link for TS0).

Control multiframe, 51 frames, 235 ms. (a) forward link on TS0, (b) reverse link on TS0. SST Ch 8, Fig 11.8

Control multiframe 2

- **Forward**: F (FCCH) then S (SCH) repeat **every 10 frames**; between them sit B (BCCH) and C (PCH / AGCH) bursts, with one idle frame.
- **Reverse**: all 51 frames are R, reverse RACH bursts. **The uplink control channel is pure random access**, so no structure is possible.
- Both live in **TS0** of the BCH ARFCN.

8.5 GSM logical channels: traffic and control the most repeated question in this chapter

Explain the working of all traffic and control channels used in GSM 4 5 6 8 4+4 2+4

TOP 9/23 81 Bh 8 (control channels), 82 Ba 2+4 (types + purposes), 80 Bh 4+4 (BCH + CCCH), 79 Ch 5 (BCCH only), 78 Ch 8, 77 Ch 6, 75 Bh 8, 72 Ash 8, 70 Bh 4 · **80 Bh asks for BCH and CCCH only**, 79 Ch for BCCH only. Read which one is wanted

Logical channels				
Traffic channels (TCHs) BTS ↔ MS		Control channels (CCHs)		
FEC-coded speech	FEC-coded data	BCHs BTS → MS	CCCHs	DCCHs BTS ↔ MS
TCH/FS 22.8 kbps	TCH/F9.6 TCH/F4.8 TCH/F2.4 22.8 kbps	BCCH	PCH BTS → MS	SDCCH
TCH/HS 11.4 kbps	TCH/H4.8 TCH/H2.4 11.4 kbps	FCCH	RACH BTS ← MS	SACCH
		SCH	AGCH BTS → MS	FACCH

Draw this tree first, then write one line per leaf. SST Ch 8

Physical against logical

- **Physical channel** = one ARFCN + one time slot number. Hardware.
- **Logical channel** = a stream of a particular kind, **mapped onto** a physical channel. Meaning.
- Many logical channels time-share the same physical channel over the multiframe.

Traffic channels (TCH) 2

- Carry **digitally encoded user speech or user data**, both directions.
- **Full rate TCH/FS**: a slot in **every** frame → 22.8 kbps.
- **Half rate TCH/HS**: a slot in **every second** frame → 11.4 kbps, doubling the users per carrier.
- Data variants: TCH/F9.6, F4.8, F2.4 and TCH/H4.8, H2.4.

1. Broadcast channels (BCH) forward only, on TS0 of the BCH ARFCN 4

- **BCCH**, broadcast control channel.
 - Broadcasts **cell and network identity**, the list of channels in use, and the access parameters.
 - **This is what a nearby mobile reads to identify a cell and lock on to it.**
- **FCCH**, frequency correction channel.
 - Occupies TS0 of the first frame, **repeated every 10th frame** of the control multiframe.
 - Lets each subscriber unit **synchronise its local oscillator** to the base station's exact frequency.
- **SCH**, synchronisation channel.
 - Broadcast in TS0 **immediately after an FCCH frame**.
 - Identifies the serving base station and gives **frame synchronisation**; the base station also issues **timing advance** here.

2. Common control channels (CCCH) call establishment 4

- Occupy TSO of every control frame **not** used by the BCH or the idle frame.
- **PCH**, paging channel, *forward*: notifies a specific mobile of an **incoming call**.
- **RACH**, random access channel, *reverse*: the mobile **acknowledges a page or originates a call**. **Contended, so collisions are possible** and a random backoff is used.
- **AGCH**, access grant channel, *forward*: the base station's answer to a RACH. Assigns a time slot, radio channel and dedicated control channel. **AGCH is the last CCCH message before the mobile leaves the control channel**.

3. Dedicated control channels (DCCH) both directions 4

- Bidirectional, same format and function each way. May sit in **any slot on any ARFCN except TSO of the BCH ARFCN**.
- **SDCCH**, stand-alone dedicated control channel.
 - Signalling during idle periods and **during call setup, before a TCH is allocated**; released as soon as the TCH is assigned.
 - Carries authentication, ciphering setup, location update and SMS.
 - **The intermediate channel that keeps mobile and base connected while resources are being allocated**.
- **SACCH**, slow associated control channel.
 - **Always paired with a TCH or an SDCCH**; occupies **every 13th frame** of the traffic multiframe.
 - *Forward*: power-level and timing-advance commands.
 - *Reverse*: received signal strength and quality reports for the serving cell **and its neighbours** → this is what feeds MAHO.
- **FACCH**, fast associated control channel.
 - Carries **urgent** signalling, above all **handover** commands.
 - Has no slots of its own: it **steals a TCH burst**, flagged by the two stealing bits in the normal burst.
 - **Far faster than SACCH**, at the cost of 20 ms of speech, which the speech decoder conceals.

8.6 Example of a GSM call, and location updating

GSM frame structure; how traffic and control channels are used while making a call 4+4 4+6

HOT 2/23 70 Bb 4+4, 77 Ch 4+6 · **the frame-structure half is \$8.4**; this band answers the second half · the channel names must come out in the right order, that is what is being marked

Mobile-originated call setup, in order

1. MS reads **FCCH** → pulls its oscillator onto the base station.
2. MS reads **SCH** → frame synchronisation and base station identity.
3. MS reads **BCCH** → cell and network parameters. It is now **camped on the correct BCH**.
4. To originate, MS sends a burst of **RACH** data on the **same ARFCN** it is locked to.
5. Base station answers with an **AGCH** message on the CCCH, assigning an **SDCCH**.

Mobile-terminated differs in one step: the network first pages the mobile on the PCH, the mobile answers on the RACH, and everything from step 6 is identical.

verified/added

Table 2.1 Location updating

System activity	Channel	Mobile activity
System overhead parameters and other overhead messages	BCCH	Mobile switched on, searches for base channel and synchronises. Monitor BCCH for current location.
Receive channel request	RACH	If current location different from that stored in SIM generate a channel request.
Assign stand alone dedicated control channel	AGCH	Receive stand alone dedicated control channel assignment and store in memory.
Receive location updating request.	SDCCH	Request for location updating (registration).
Request authentication from mobile	SDCCH	Receive authentication request.
Receive and check authentication	SDCCH	Authentication response
Request mobile to transmit in ciphered mode	SDCCH	Receive request and switch to ciphered mode.
Receive acknowledgement	SDCCH	Acknowledge cipher mode request.
Confirm location updating including the optional assignment of a temporary identity (TMSI).	SDCCH	Receive location updating including and (TMSI) and store in SIM.
Receive acknowledgement	SDCCH	Acknowledge new location and TMSI.
Send channel release	SDCCH	Switch to idle update mode, monitor BCCH and CCCH.

Location updating, channel by channel: which logical channel carries each step when a mobile is switched on in a new area. Note how much of it rides on the SDCCH. SST Ch 8, Table 2.1

6. MS, which is monitoring **slot 0 of the BCH**, reads its ARFCN and slot assignment from the AGCH and **retunes to the SDCCH**.
7. Over the SDCCH the MS receives **timing advance** and **power level** commands. Only now can it send proper normal bursts.
8. SDCCH carries **authentication** messages both ways. Meanwhile the PSTN connects the dialled party to the MSC, and the MSC switches the speech path to the serving base station.
9. Base station commands the MS over the SDCCH to a **new ARFCN and slot for the TCH**.
10. MS tunes to the **TCH**; speech flows both ways and the **SDCCH is vacated**.

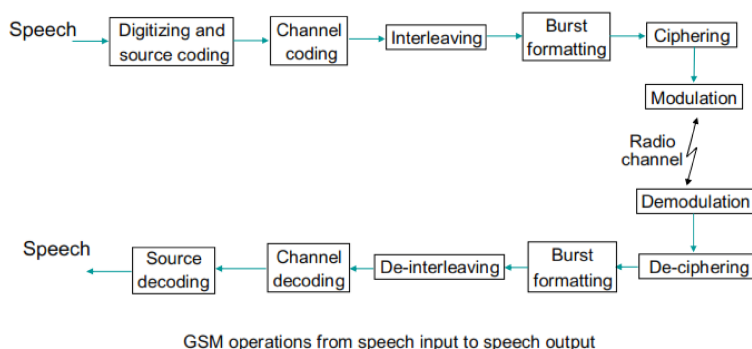
What each stage costs, and why it is built this way

- Steps 1–3 are **listen only**. The mobile transmits nothing until it has read the BCH, which is why a handset can be switched on inside a foreign network safely.
- Step 4 is the **only contended transmission** in the whole sequence, so it uses the short RACH burst with its extended guard period (§8.4).
- Step 7 exists because the mobile does not yet know its distance from the base station. **Without timing advance its normal burst would overlap the next slot.**
- The SDCCH does **all the work of call setup** and is then thrown away. That is what “stand-alone dedicated” means: dedicated to one mobile, standing alone from any traffic channel.
- Once on the TCH, further signalling has to **steal bursts (FACCH)** or wait for the 13th frame (SACCH), because there is no spare capacity left.

8.7 Signal processing in GSM: speech in, radio out

Basic signal processing operations to convert speech to a radio signal and back 8

PIN 2/23 71 Bh 8, 81 Ba 8 · **always the full 8 marks** · speech coding itself is Ch 6, channel coding and interleaving are Ch 6, equalization is Ch 5. Here you assemble them into one chain



Draw the forward chain, then say the reverse is its mirror. The radio channel sits between modulation and demodulation. Rappaport Fig 10.11, via SST Ch 8

The chain, block by block

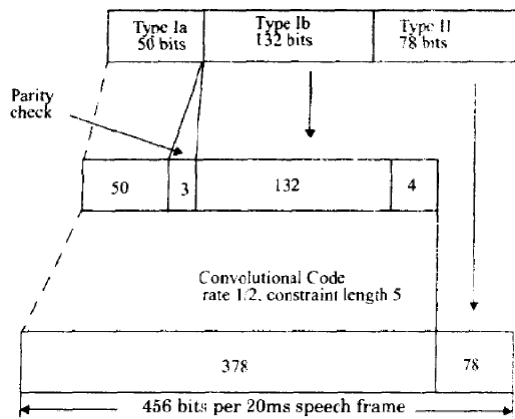
1. **Digitizing and source coding**: RPE-LTP speech coder, **260 bits per 20 ms = 13 kbps**.
2. **Channel coding**: adds redundancy, **260 → 456 bits = 22.8 kbps**.
3. **Interleaving**: the 456 bits are cut into **eight 57-bit sub-blocks** and spread over **8 consecutive TCH slots**.
4. **Burst formatting**: adds tail, training, stealing and guard bits → the 156.25-bit burst of §8.4.
5. **Ciphering**: A5 stream cipher, keystream XORed with the data bits.
6. **Modulation**: **0.3 GMSK at 270.833 kbps**.
7. Radio channel, then **demodulation, de-ciphering, burst formatting, de-interleaving, channel decoding, source decoding** → speech.

Speech coding 2

- Coder is a **REL P / RPE-LTP**: residually excited linear predictor with a long term predictor. Full detail in Ch 6.
- Output **260 bits per 20 ms frame**. Half-rate codec runs at 6.5 kbps.
- **VAD** = voice activity detector. Speech is active only about **40 %** of the time, so **DTX** switches the transmitter off in the gaps.
- **DTX saves handset battery and cuts interference to other cells**. Comfort noise is inserted at the far end so the line does not sound dead.

Channel coding 3

- The 260 bits are **sorted by how much they matter to speech quality**:
 - **50 type Ia** bits, most important → **3 CRC parity** bits added.
 - **132 type Ib** bits, next.
 - **78 type II** bits, least important → **no protection at all**.
- Types Ia and Ib + parity + **4 tail zeros** = 189 bits go through a **rate 1/2, constraint length $K = 5$** convolutional encoder → 378 bits.
- $378 + 78 = 456$ bits per 20 ms = **22.8 kbps**.
- **Unequal error protection**: spending all the redundancy on the bits that would wreck the speech is what makes 13 kbps survive a fading channel.



260 bits in, 456 bits out. SST Ch 8

Interleaving, and why 8 slots

- A deep fade wipes out a **burst** of adjacent bits, which no convolutional code handles well.
- Spreading one speech frame over **8 bursts in 8 different frames** turns one long error burst into eight short ones, **one per code frame**, which the decoder can fix.
- Cost: **about 40 ms of extra delay**, which is why the specification table lists interleaving depth as a delay figure.
- **This is why a GSM call survives a moving car.** The channel decorrelates between bursts, so eight bursts are eight independent looks at the same speech frame, i.e. **time diversity** (Ch 5).

8.8 CDMA standards: IS-95 frequency and channel specification

Frequency and channel specification of the IS-95 CDMA standard 2 4

no paper asks this alone, but every IS-95 channel question opens with it · the CDMA *principle*, near-far effect, encoding and decoding are all Ch 7; this chapter is only the *standard*

IS-95, also called cdmaOne

- A **US standard** built on Qualcomm's technology, the first commercial CDMA cellular system.
- **FDD**, using two **1.25 MHz simplex channels 45 MHz apart**.
 - Reverse (uplink) **824–849 MHz**.
 - Forward (downlink) **869–894 MHz**.
- Chip rate **1.2288 Mcps**, i.e. 128×9.6 kbps.
- **Users in a cell and users in adjacent cells share the same radio channel**, so the frequency reuse factor is $N = 1$ and **no frequency planning is needed**. This is the single biggest planning advantage over GSM.
- $N = 1$ is also what makes **soft handoff** possible: the mobile can be on the same carrier with two base stations at once (Ch 2).
- Variable-rate vocoder: **9.6 / 4.8 / 2.4 / 1.2 kbps**. Later **EVRC** gives 13 kbps quality at the 8 kbps rate.

The three code families 3

- **Walsh codes**, 64 rows of a 64×64 Hadamard matrix.
 - *Forward link*: one Walsh code per **logical channel**, so they separate **channels**.
 - *Reverse link*: 64-ary orthogonal modulation, so they separate **groups of six symbols**, not users. **Learn this asymmetry, it is examinable.**
- **Short PN codes**, a quadrature pair of period 2^{15} chips.
 - Applied on the I and Q arms after Walsh spreading.
 - Each cell or sector uses one of **512 phase offsets**, so the short code **identifies the sector**.
- **Long PN code**, period $2^{42} - 1$, generated from a 42-bit mask.
 - *Forward*: scrambles and encrypts the traffic, mask derived from the user's **ESN**.
 - *Reverse*: **it is the channelising code** — each mobile is separated by its own long-code phase.

Power control, because CDMA lives or dies by it 2

- **Open loop**: the mobile measures the received **pilot** strength and sets its own transmit power inversely.
- **Closed loop**: the base station measures each mobile and sends **power control bits** at **800 Hz**, ± 1 dB per bit.
- Without it the **near-far problem** (Ch 7 §7.6) destroys capacity.

8.9 Forward CDMA (IS-95) channel the standard 4 to 6 mark drawing

Draw and explain the forward CDMA (IS-95) channel; what is a channelization code? 4 6

TOP 7/23 80 Bb 4, 80 Ch 6, 82 Ba 6, 78 Ch 4, 73 Ma 4 (channelization code + forward channels), 73 Bb 6 (pilot and sync channels only), 81 Ba 4 (reverse, §8.10) · **73 Bb wants only the pilot and sync blocks**, drawn out in full

– Walsh code generation

$$\begin{aligned}
 H_1 &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} & H_2 &= \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\
 H_4 &= \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} & H_{2N} &= \begin{bmatrix} H_N & H_N \\ H_N & \overline{H_N} \end{bmatrix}
 \end{aligned}$$

where N is a power of 2.

The n th row of the Walsh matrix represents Walsh code n (W_n), where n starts from 0.

Channelization code = Walsh code 2, built by the Hadamard recursion. Row n of the matrix is W_n . Worked as a numerical in Ch 7 companion §4. SST Ch 8

Why Walsh codes work here

- All 64 rows are **mutually orthogonal**: map $0 \rightarrow -1$, $1 \rightarrow +1$ and the term-by-term product of any two rows sums to 0.
 - So a receiver that correlates with W_n recovers channel n and sees **exactly zero** from the other 63, provided they arrive in step.
 - They stay in step on the forward link because **one base station transmits them all**. That is why the reverse link cannot use them for channelisation.
3. **Paging channels**, W_1 to W_7 , up to seven.
- **4.8 or 9.6 kbps**, the rate announced by the sync message.
 - Overhead messages, **pages**, and **traffic channel assignment** during call setup.
 - A mobile monitors **one** paging channel, and ignores it once a traffic channel is up.

The four forward logical channels 4

1. Pilot channel, W_0 .

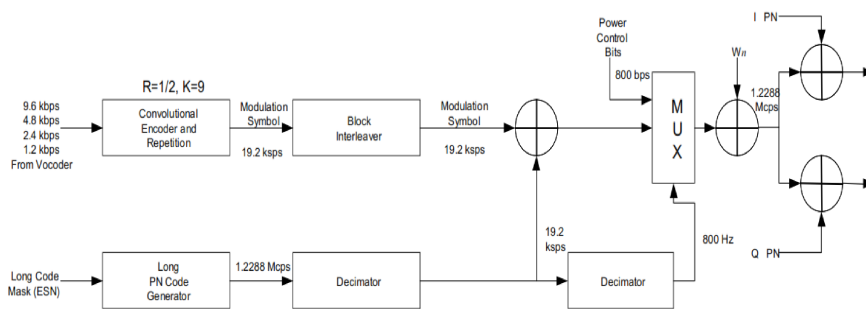
- Carries **no baseband information at all**: an all-zero stream, spread by W_0 which is itself all zeros, so what goes out is **the short PN pair itself** at the sector's offset.
- Transmitted **continuously**. Gives the mobile **timing acquisition and a phase reference** for coherent detection.
- Its E_c/I_0 is what the mobile measures to decide **which sector is strongest**, i.e. it drives handoff.

2. Sync channel, W_{32} .

- **1.2 kbps** of real baseband data, convolutionally coded and interleaved.
- Carries the sync channel message: **system time, long-code state, the sector's PN offset, and the paging channel rate (PRAT)**.
- Uses the **same PN offset as the pilot**, so once the pilot is acquired the sync channel is readable at once.

4. Forward traffic channels, the remaining Walsh codes.

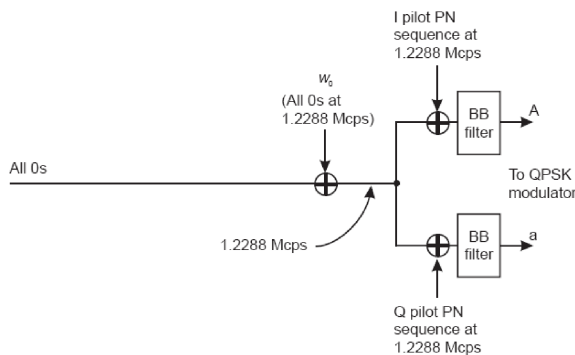
- User voice and data at **1.2 / 2.4 / 4.8 / 9.6 kbps**.
- Structure is the paging channel plus one addition: **multiplexed power control bits at 800 bps**, punctured into the stream.
- Scrambled by the long code with that user's **ESN-derived mask**.



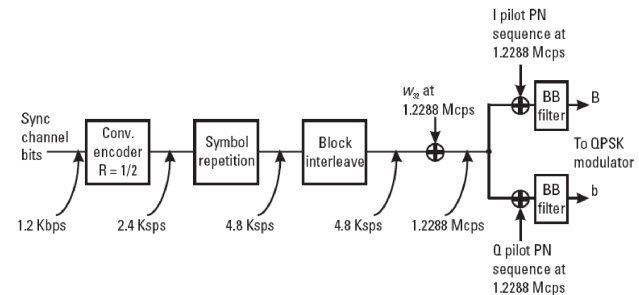
Forward traffic channel, the block diagram to reproduce. Rate $1/2$ $K = 9$ convolutional encoder $\rightarrow$ symbol repetition to 19.2 kbps $\rightarrow$ block interleaver $\rightarrow$ long-code scrambling $\rightarrow$ MUX with the 800 bps power control bits $\rightarrow$ Walsh cover $W_i \rightarrow$ I/Q short PN $\rightarrow$ QPSK. SST Ch 8

The modulator, and how the channels combine 2

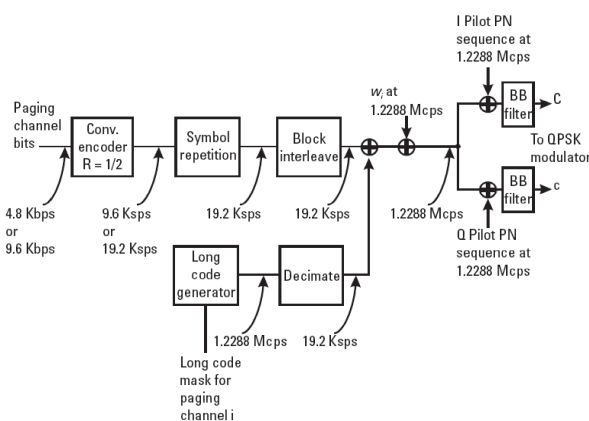
- Every logical channel first passes a **gain control** block, which sets how much of the sector's power that channel gets.
 - Traffic channel gains **change continuously**, driven by forward power control.
 - Pilot gain is fixed and deliberately generous, typically 15 to 20 % of total power, because everything else depends on it. **verified/added**
- After gain setting all channels are **coherently added** into one composite spread-spectrum signal.
- The sum is split into **I and Q**, each spread by its short PN sequence, up-converted on quadrature carriers and added $\rightarrow$ one **QPSK** passband signal.
- One 1.25 MHz carrier therefore carries all 64 code channels of the sector at once.**



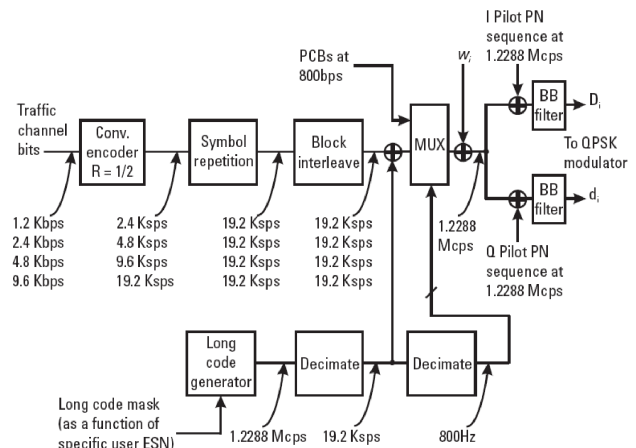
Pilot channel 3. All zeros in, PN out. No encoder, no interleaver, nothing to decode. SSD, IS-95 channels



Sync channel 3. 1.2 kbps $\rightarrow$ rate $1/2 \rightarrow$ 2.4 kbps $\rightarrow$ repetition to 4.8 kbps $\rightarrow$ interleave $\rightarrow$ $W_{32} \rightarrow$ I/Q PN. SSD, IS-95 channels



Paging channel. Same shape as the sync channel plus long-code scrambling through a decimator. SSD, IS-95 channels



Forward traffic channel. The paging channel plus the **MUX that punctures in the 800 bps power control bits**. SSD, IS-95 channels

8.10 Reverse CDMA (IS-95) channel

Draw the reverse CDMA (IS-95) channel 4

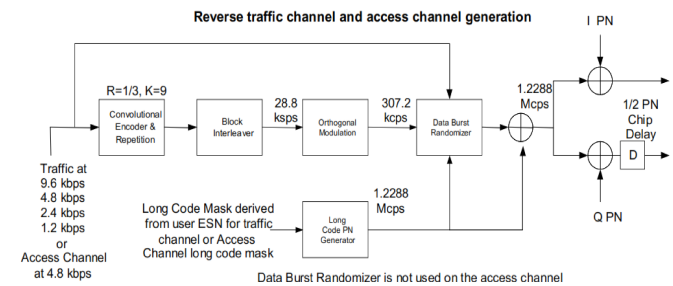
81 Ba 4 · counted inside the TOP 6/23 forward/reverse total of \$8.9 · the marks are usually won on **why the reverse link is built differently**, not on the drawing

Two channels only, and no pilot

- The reverse link supports **access channels** and **traffic channels**. That is all.
- **There is no reverse pilot** in IS-95: it is impractical for every mobile to broadcast its own pilot sequence.
- No pilot → no phase reference → the reverse link is **non-coherent**.
- Non-coherent → **Walsh functions cannot channelise**, because mobiles cannot be held chip-aligned with each other.
- Instead **each mobile is identified by its own long PN sequence**, set by a mask derived from its ESN.
- Walsh functions are still used, but for **64-ary orthogonal modulation**: every six code symbols are sent as one of the 64 Walsh functions, and the base station decides which one by correlating against all 64.
- **Forward: Walsh separates channels. Reverse: Walsh represents symbols.**

1. Access channel

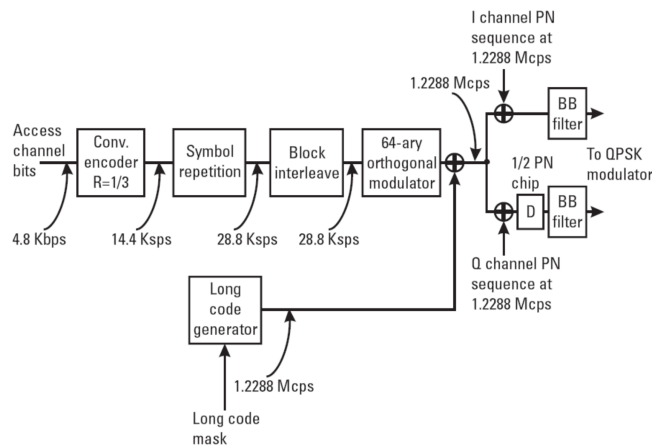
- Used when the mobile **has no traffic channel assigned**: to register, originate calls, respond to pages and orders, and send overhead messages.
- Baseband rate is **fixed at 4.8 kbps**.
- **Rate 1/3 convolutional encoder**, not 1/2 as on the forward link. The reverse is the weaker of the two links (a handset transmits under a watt), so it is given more redundancy.
- Symbol repetition doubles it to **28.8 kbps**, then block interleaving, then the 64-ary orthogonal modulator.



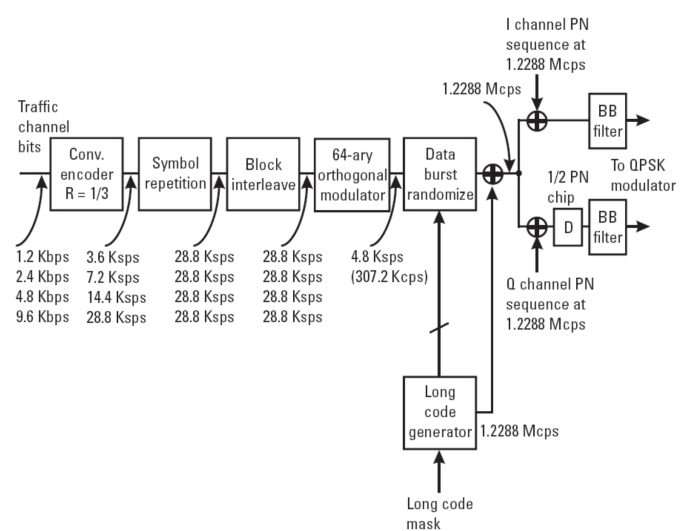
Reverse traffic and access channel generation. Rate $1/3 K = 9$ encoder → repetition to 28.8 kbps → interleaver → 64-ary orthogonal modulator, 307.2 kcps → data burst randomizer → long PN → I/Q short PN with a **half-chip delay on Q**, giving OQPSK. SST Ch 8

2. Reverse traffic channel

- Carries user voice, data and signalling at **1.2 / 2.4 / 4.8 / 9.6 kbps**.
- Structure is the access channel plus one block: the **data burst randomizer**.
- **The randomizer is the whole point**. When the vocoder drops to a lower rate the repeated symbols are redundant, so the randomizer **gates them off** instead of transmitting them.
- Result: the handset transmits only in pseudo-random bursts → lower average power, longer battery life and **less interference into every other user**, which is capacity.
- The forward link exploits the same voice activity differently: it **lowers the energy per symbol** rather than gating off, since the base station must keep the composite waveform continuous.



Access channel 2. Note the fixed 4.8 kbps input and the absence of a data burst randomizer. SSD, IS-95 channels



Reverse traffic channel 2. The one added block is the **data burst randomizer**. SSD, IS-95 channels

8.11 GSM against CDMA, and CDMA against LTE

Compare GSM and CDMA standards 3 4

76 Bh 3, 71 Ma 4 · 71 Ma phrases it as “**what is GSM and CDMA standard**” before asking for the GSM architecture, so the comparison is the short half

Point	GSM	CDMA (IS-95)
Multiple access	TDMA over FDMA	CDMA, direct sequence
Duplexing	FDD	FDD
Carrier bandwidth	200 kHz	1.25 MHz
Users per carrier	8 (16 at half rate)	about 55 to 61 code channels
Frequency reuse	$N = 3, 4, 7, 9$; needs planning	$N = 1$, no planning
Modulation	0.3 GMSK	QPSK forward, OQPSK reverse
Speech coder	RPE-LTP 13 kbps, fixed	variable rate 9.6 down to 1.2 kbps
Handoff	hard (break before make)	soft and softer (make before break)
Power control	slow, on SACCH	fast, 800 Hz closed loop , essential
Equalizer	required , 26-bit training sequence	not needed; RAKE receiver instead
Capacity limit	hard: run out of slots	soft: interference limited , degrades gently
Subscriber identity	SIM card , moves between handsets	tied to the handset ESN
Security	A3/A5/A8 over the air	inherent in the long PN code, plus ciphering

Compare the system architecture of CDMA with LTE; mention the function of the entities 3+5

PIN 1/23 74 Ma 3+5 · verified/added no lecture deck covers LTE at all · 8 marks, so it is worth learning the seven LTE boxes by name

CDMA (IS-95 / cdma2000) architecture

- MS → BTS → BSC → MSC for voice, or PDSN for packet data.
- BTS: radio, Walsh covering, power control commands.
- BSC: radio resource control, soft handoff combining, and the vocoders.
- MSC: circuit switching to the PSTN.
- PCF + PDSN: the packet data path, bolted on later.
- HLR / VLR / AuC: same roles as in GSM.
- Hierarchical and circuit-first: voice and data ride separate cores.

LTE (EPS) architecture

- UE → eNodeB → EPC (MME, S-GW, P-GW) → external IP networks.
- eNodeB: radio plus all the radio resource control that a BSC used to do. There is no controller node. eNodeBs talk to each other directly over X2.
- MME, control plane only: attach, authentication, bearer setup, mobility and paging.
- S-GW: user-plane anchor, forwards packets during inter-eNodeB handover.
- P-GW: IP anchor to the outside world, allocates the IP address, enforces policy, does charging.
- HSS: subscriber database, HLR + AuC in one.
- PCRF: policy and charging rules.
- Flat and all-IP: two nodes from handset to internet, and no circuit domain at all.

The three differences to write down 3

- **Topology**: CDMA is BTS-BSC-MSC, three tiers. LTE is eNodeB-EPC, two tiers. The BSC's job moved into the base station, which cuts handover latency.
- **Bearer**: CDMA is circuit-switched voice with packet data added later. LTE is all-IP, packet only; voice comes back as VoLTE over IMS.
- **Air interface**: 1.25 MHz DS-CDMA against OFDMA down, SC-FDMA up, with MIMO, on a scalable 1.4 to 20 MHz carrier. Peak rate rises from about 3 Mbps to 100 Mbps and beyond.

8.12 Evolution of wireless telephone systems syllabus, no PYQ yet

AMPS, CT2, DECT, PHS, PACS, IS-54 / IS-136, IS-94, IS-95: the standards named in the syllabus

syllabus, no PYQ yet · syllabus item 8(a), never examined in 23 papers · the *generations* question (1G to 5G) is Ch 1 §1.3 and is asked every year; this band is the *systems* behind those generations · read the three tables, do not memorise them

First generation, analog cellular

- All FM voice with FSK signalling, FDMA/FDD, one carrier per user for the whole call.
- AMPS, the Americas: 824–849 / 869–894 MHz, 30 kHz channels, 832 channels.
- TACS / ETACS, UK and Europe: 25 kHz channels, up to 1240 channels.
- NMT-450 / NMT-900, the Nordic countries, 25 and 12.5 kHz.
- C-450 Germany, RTMS Italy, Radiocom 2000 France, NTT and JTACS/NTACS Japan.
- Every country built its own, which is exactly the fragmentation GSM was created to end.

Table 10.10 Analog Cellular Systems Overview

Standard	Mobile Tx/Base Tx (MHz)	Channel spacing (kHz)	Number of Channels	Region
AMPS	824–849/869–894	30	832	The Americas
TACS	890–915/935–960	25	1000	Europe
ETACS	872–905/917–950	25	1240	United Kingdom
NMT 450	453–457.5/463–467.5	25	180	Europe
NMT 900	890–915/935–960	12.5	1999	Europe
C-450	450–455.74/460–465.74	10	573	Germany, Portugal
RTMS	450–455/460–465	25	200	Italy
Radiocom 2000	192.5–199.5/200.5–207.5 215.5–233.5/207.5–215.5 165.2–168.4/169.8–173 414.8–418/424.8–428	12.5	560 640 256 256	France
NTT	925–940/870–885 915–918.5/860–863.5 922–925/867–870	25/6.25 6.25 6.25	600/2400 560 480	Japan
JTACS/NTACS	915–925/860–870 898–901/843–846 918.5–922/863.5–867	25/12.5 25/12.5 12.5	400/800 120/240 280	Japan

Rappaport Table 10.10

Table 10.11 Digital Cordless Air Interface Parameters Summary

	CT2	CT2+	DECT	PHS	PACS
Region	Europe	Canada	Europe	Japan	U.S.A.
Duplexing	TDD		TDD	TDD	FDD or TDD
Frequency band (MHz)	864–868 944–948		1880–1900	1895–1918	1850–1910/ 1930–1990 or 1920–1930
Carrier spacing (kHz)	100		1728	300	300/300
Number of carriers	40		10	77	400 or 32
Channels/carrier	1		12	4	8 or 4
Channel bit rate kbps	72		1152	384	384
Modulation	GFSK		GFSK	$\pi/4$ DQPSK	$\pi/4$ QPSK
Speech Coding	32 kb/s		32 kb/s	32 kb/s	32 kb/s
Average handset TX power (mW)	5		10	10	25
Peak handset TX power (mW)	10		250	80	100
Frame duration (ms)	2		10	5	2.5 or 2.0

Digital cordless standards. Rappaport Table 10.11

Cordless and PCS, the short-range family

- **CT2**, Europe and Canada, 864–868 MHz, **TDD**, 100 kHz carriers, **one channel per carrier**, 72 kbps GFSK, ADPCM speech.
- **DECT**, Europe, 1880–1900 MHz, TDD, **1728 kHz carriers with 12 channels each**, 1152 kbps GFSK. The one that survived, as the office and home cordless phone.
- **PHS**, Japan, 1895–1918 MHz, TDD, 300 kHz, 4 channels per carrier, 384 kbps $\pi/4$ -DQPSK.
- **PACS**, USA, **FDD or TDD**, 300 kHz, 8 or 4 channels per carrier, 384 kbps $\pi/4$ -QPSK. Designed for licensed PCS microcells.
- **IS-94** is the odd one out: it is not cordless but the standard for **AMPS-based private and in-building microcell systems**, i.e. cellular hardware reused as a premises system. verified/added
- **All of them are TDD, low power and short range**, which is what separates a cordless standard from a cellular one.

Table 10.12 Second Generation Digital Cellular Standards Summary

	GSM	IS-54	PDC
Year of Introduction	1990	1991	1993
Frequencies	890–915 MHz (R) 935–960 MHz (F)	824–849 MHz (R) 869–894 MHz (F)	810–830 & 1429– 1453 MHz (R) 940–960 & 1477– 1501 MHz (F)
Multiple Access	TDMA/FDMA/ FDD	TDMA/FDMA/ FDD	TDMA/FDMA/ FDD
Modulation	GMSK ($BT=0.3$)	$\pi/4$ DQPSK	$\pi/4$ DQPSK
Carrier Separation	200 kHz	30 kHz	25 kHz
Channel Data Rate	270.833 kbps	48.6 kbps	42 kbps
Number of Voice Channels	1000	2500	3000
Spectrum Efficiency	1.35 bps/Hz	1.62 bps/Hz	1.68 bps/Hz
Speech Coding	REL P-LTP @ 13 kbps	VSELP @ 7.95 kbps	VSELP @ 6.7 kbps
Channel Coding	CRC with $r=1/2$; $L=5$ Conv.	7 bit CRC with $r=1/2$; $L=6$ Conv.	CRC with Conv.
Equalizers	Adaptive	Adaptive	Adaptive
Portable Tx. Power Max./Avg.	1 W/125 mW	600 mW/200 mW	

Second generation digital cellular. Rappaport Table 10.12

Second generation, digital cellular

- **GSM (1990)**: TDMA/FDMA/FDD, 200 kHz, GMSK $BT = 0.3$, 270.833 kbps, RPE-LTP at 13 kbps, **1.35 bps/Hz**.
- **IS-54**, later **IS-136**, also called **USDC** or **D-AMPS** (1991): **30 kHz channels, deliberately the same as AMPS** so the two could share a band, 3 users per carrier, $\pi/4$ -DQPSK at 48.6 kbps, VSELP 7.95 kbps.
- **PDC**, Japan (1993): 25 kHz, $\pi/4$ -DQPSK at 42 kbps, VSELP 6.7 kbps, **1.68 bps/Hz**, the most spectrally efficient of the three.
- **IS-95 (1993)**: the CDMA outlier, 1.25 MHz carriers, $N = 1$. §8.8.
- **Three of the four are TDMA**. CDMA was the minority choice in 2G and the majority choice in 3G.

8.13 WiFi, WiMAX, UMTS, EVDO, UMB, LTE and recent trends no lecture note covers this

Write short notes on WiMAX / LTE 3 4 5

HOT 3/23 78 Ch 4 (WiMAX), 73 Ma 5 (WiMAX), 72 Ash 3 (WiMAX) + 3 (LTE) · verified/added **absent from all three lecture decks** · WiFi and WLAN are answered in Ch 7 §7.11, not here

WiMAX 4

- **WiMAX** = worldwide interoperability for microwave access, the trade name for **IEEE 802.16**. A **WMAN**, one step above WiFi's WLAN.
- Two versions worth naming:
 - **802.16d** (2004), **fixed** WiMAX, OFDM.
 - **802.16e** (2005), **mobile** WiMAX, **scalable OFDMA**, supports handover at vehicular speed.
- Bands: **2–11 GHz for NLOS**, 10–66 GHz for LOS backhaul. Scalable channel bandwidth 1.25 to 20 MHz.
- Throughput about **75 Mbps fixed**, 15 Mbps mobile; range 5–8 km NLOS, up to 50 km LOS.
- **The MAC is the real difference from WiFi**: connection-oriented and **scheduled by the base station**, not contention. Every subscriber gets a grant, so QoS classes (UGS, rtPS, nrtPS, best effort) are enforceable.
- Topologies: point-to-multipoint, and mesh.
- **Uses**: last-mile broadband where cable is uneconomic, backhaul for WiFi hotspots and cell sites, rural and disaster-recovery connectivity.
- (-): needs licensed spectrum, expensive base stations, and it **lost the 4G market to LTE** because operators already owned 3GPP cores.

Recent trends 2

- **5G NR**: sub-6 GHz plus **mmWave**, massive MIMO and beamforming, network slicing, three service classes **eMBB** / **URLLC** / **mMTC**.
- **IoT**: NB-IoT and LTE-M, narrowband, low power, huge device counts.
- **D2D** device-to-device, **C-RAN** centralised baseband, edge computing.
- **Cross-refer Ch 1 §1.4**, which answers the “trends beyond 3G” question in full. Do not repeat it here.

LTE 3

- **LTE** = long term evolution, 3GPP Release 8 (2008). The 4G standard that won.
- **OFDMA** on the downlink, **SC-FDMA** on the uplink. **SC-FDMA is chosen purely for its low PAPR**, which is what a battery-powered handset amplifier needs (Ch 4 §4.6).
- **MIMO** up to 4×4 , scalable bandwidth **1.4 / 3 / 5 / 10 / 15 / 20 MHz**, both FDD and TDD.
- Release 8 peaks at about **300 Mbps down, 75 Mbps up**.
- **Flat all-IP** architecture, §8.11. **No circuit-switched domain**: voice is VoLTE over IMS.
- **LTE-Advanced** (Rel 10) adds **carrier aggregation**, 8×8 MIMO, relay nodes and CoMP, reaching **1 Gbps**.

The other names on the syllabus 2

- **UMTS**: 3GPP's 3G. **W-CDMA**, **5 MHz carriers, 3.84 Mcps**. Radio network is NodeB + RNC, reusing the GSM/GPRS core. HSPA later takes it to 14.4 Mbps.
- **CDMA-EVDO**: cdma2000 **1xEV-DO**, “**evolution, data optimised**”. A *separate* 1.25 MHz data-only carrier with a time-multiplexed forward link and rate adaptation. Rev 0 gives 2.4 Mbps, Rev A 3.1 Mbps down / 1.8 Mbps up.
- **UMB**: ultra mobile broadband, 3GPP2's OFDMA answer to LTE. **Abandoned in 2008** when Qualcomm stopped development and the CDMA operators moved to LTE. Worth one honest line: it is on the syllabus but was never deployed. verified/added

	WiFi	WiMAX	LTE
Standard	802.11	802.16	3GPP Rel 8
Scale	WLAN, 100 m	WMAN, km	cellular WAN
Spectrum	unlicensed	licensed	licensed
MAC	contended, CSMA/CA	scheduled	scheduled
QoS	weak	built in	built in
Mobility	nomadic	vehicular (802.16e)	full handover

If a question asks you to “compare”, this table is the answer.

The row that actually decides all the others is **contended** against **scheduled**. verified/added

8.14 Regulatory issues the highest-frequency topic in this chapter

Describe regulatory issues in wireless communication: spectrum allocation, pricing, licensing, tariffs, interconnection 4 5

TOP 9/23 81 Bh 4 (licensing + allocation), 80 Ba 4, 79 Bh 4 (spectrum regulations), 76 Bh 5, 74 Bh 5, 73 Bh 5 (spectral licensing), 80 Ch 2+4, 79 Ch 2+3, 70 Ma 4 · 80 Ch and 79 Ch ask "significance of spectrum management" first 2, then the functions of the regulator

Why regulation exists at all 2

- Radio spectrum is a **finite, non-storable, shared** natural resource.
- Interference is **non-excludable**: two users on the same frequency, in the same place, at the same time both fail. Neither can protect itself.
- So a government must decide **who may use which band, where, and on what terms**. That is spectrum management.
- **Significance of spectrum management** 2 :
 - Interference-free coexistence of services.
 - **Efficient use** of a scarce resource, and no hoarding.
 - **International harmonisation** → roaming, economies of scale in handsets.
 - Protected bands for **aviation, defence, emergency and scientific** use.
 - Predictable rules → operators will invest.

1. Spectrum allocation

- ITU-R divides the world into **three Regions** and allocates bands to *services* at the World Radiocommunication Conference.
- The **national regulator** then assigns those bands to *operators*: NTA in Nepal, FCC in the USA, CEPT/ETSI in Europe.
- Assignment methods: **administrative** (a "beauty contest" on merit), **auction**, lottery, first-come first-served, or left **unlicensed as a commons** (the ISM bands WiFi and Bluetooth live in).

2. Spectrum pricing

- Charges an operator for holding spectrum: an **annual royalty** per MHz per head of population, or the one-off **auction price**.
- Aim: **make idle spectrum expensive**, so it moves to whoever will use it best.
- **Risk**: overpricing is passed straight to the subscriber as tariff, and slows rural rollout. **The regulator has to balance revenue against coverage**.

4. Tariff regulation

- Regulator **approves tariffs**, and classifies them as **fixed, maximum, or non-regulated**.
- Price caps and floors protect subscribers from a dominant operator and rivals from **predatory pricing**.
- Tariffs are usually deregulated as competition arrives.

3. Licensing

- Decides **who may operate**, over what area, for how long, and on renewal terms.
- Attaches obligations: **coverage targets, quality of service, rural rollout**, lawful interception.
- **Spectrum caps** stop one operator cornering the band and keep the market competitive.
- **Type approval**: only equipment that meets emission limits may be connected.
- Technology-neutral licences let an operator reform a 2G band for LTE without a new licence.

5. Interconnection

- Every licensee must be able to **terminate calls on every other network**, otherwise the first operator to scale wins by default.
- Regulator sets **cost-oriented interconnection charges**, the points of interconnection, and **number portability**.
- It also **settles interconnection disputes**, which is why dispute resolution appears in the list of functions below.

Functions of the regulator 4 the list 80 Ch and 79 Ch want

1. **Inspection and investigation** of the affairs of telecom operators and internet service providers.
2. **Settle disputes** between licensees, or between a licensee and a customer, relating to telecommunications service.
3. **Determine the quality and standard** of the machinery, equipment and facilities used, and of the service itself.
4. **Licensing** of telecom operators and internet service providers.
5. **Approval of tariffs**: fixed tariffs, maximum tariffs and non-regulated tariffs.

SST Ch 8, the NTA regulatory approach · write these five and the 4 marks are **secure**. Add spectrum allocation and monitoring as a sixth if the question says "spectrum management".

Nepal, if a local example is wanted verified/added

- NTA = Nepal Telecommunications Authority, constituted under the **Telecommunications Act 2053**.
- Licence categories it issues include **basic telephony, cellular mobile, internet service, VSAT and rural VSAT, network service providers, international trunk telephone** and rural ISP.
- **Naming NTA and the Act is enough** for a Nepali paper. Do not attempt licence counts, they change every year.

Answer skeleton 5

1. One line: spectrum is finite and shared, so it must be regulated. 1
2. Name the **five heads**: allocation, pricing, licensing, tariff, interconnection. 1
3. Two sentences on each head. 2.5
4. Close with ITU-R above and the national regulator (NTA) below. 0.5

8.15 Last-minute recall for this chapter**Must memorise**

- GSM 900: **890–915** up, **935–960** down, **45 MHz** apart, 200 kHz ARFCN, **124** duplex channels, 8 users each.
- **0.3 GMSK, 270.833 kbps** per carrier, 33.854 kbps per user, 24.7 kbps max user data.
- Slot **156.25 bits, 576.9 μ s**; frame **8 slots, 4.615 ms**; bit **3.692 μ s**.
- Normal burst **3 + 57 + 1 + 26 + 1 + 57 + 3 + 8.25**.
- Multiframe **26 (traffic, 120 ms)** or **51 (control, 235 ms)**; superframe **1326 frames, 6.12 s**; hyperframe **2048 superframes, 3 h 28 min**.
- Frame efficiency **74.24 %** → Ch 7 companion §3.
- Speech **260 bits / 20 ms = 13 kbps** → coded to **456 bits = 22.8 kbps** → interleaved over **8 slots**.
- Channel coding split **50 Ia + 132 Ib + 78 II**, rate $1/2$, $K = 5$.
- Control channels: **BCH = BCCH, FCCH, SCH · CCCH = PCH, RACH, AGCH · DCCH = SDCCH, SACCH, FACCH**. SACCH every **13th** frame; FACCH **steals** a TCH burst.
- IS-95: **1.25 MHz**, 824–849 / 869–894, chip rate **1.2288 Mcps**, $N = 1$, power control at **800 Hz**.
- Forward channels: **pilot W_0 , sync W_{32} , paging $W_1–W_7$, traffic**. Reverse: **access and traffic only, no pilot**.
- Forward encoder **rate $1/2$ $K = 9$** ; reverse **rate $1/3$ $K = 9$** + 64-ary orthogonal modulation + data burst randomizer.

Most repeated, in order

1. Regulatory issues, or significance of spectrum management **TOP 9/23**
2. GSM traffic and control channels **TOP 8/23**
3. GSM system architecture, or the NSS alone **TOP 7/23**
4. Forward and reverse CDMA (IS-95) channels **TOP 6/23**
5. GSM frame structure and hierarchy **HOT 5/23**
6. WiMAX short note **HOT 3/23**
7. Signal processing in GSM **PIN 2/23** · how channels are used while making a call **HOT 2/23**
8. CDMA against LTE architecture **PIN 1/23** · GSM against CDMA, specifications of GSM, LTE short note

The best hour you can spend on this chapter is the control channel tree: nine channels, one line each, plus which direction each one points. Eight of the twenty-two papers ask for it directly, and it also supplies most of the GSM-call answer in §8.6.

No numerical companion for this chapter. No paper from 2070 to 2082 sets a calculation in Chapter 8. The one number the syllabus touches, GSM frame efficiency, is examined as a Chapter 7 multiple-access question and is worked in full in Ch 7 – Numerical Problems & Solutions, §3, including the 74 Ma "cannot reach 100 %" proof.

Two traps in this chapter: BCH is the broadcast channel here and a block code in Ch 6; and FCCH (GSM frequency correction channel) is not FCC (forward control channel, Ch 2). Read the question's context before you expand either.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver

Short	Means	Short	Means
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELTP	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access

Short	Means	Short	Means
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK, DBPSK	differential (binary) PSK
CPFSK	continuous phase FSK	MFSK, MPSK	M-ary FSK, M-ary PSK
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB, VSB	double, single, vestigial sideband
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	<i>BT</i>	GMSK filter bandwidth $\times$ symbol time

Short	Means	Short	Means
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
SSMA	spread spectrum multiple access	SDMA	space division multiple access
FHMA	frequency hopped multiple access	DSMA	direct sequence multiple access (= CDMA)
FCDMA	hybrid FDMA/CDMA	DS/FHMA	hybrid direct sequence / frequency hopped
TCDMA	time division CDMA	TDFH	time division frequency hopping
IM	intermodulation (product)	MU-MIMO	multi-user MIMO
WLAN	wireless local area network	WMAN	wireless metropolitan area network
WPAN	wireless personal area network	AP	access point
BSS, IBSS	basic service set, independent (ad hoc) BSS	ESS	extended service set
CSMA/CA	carrier sense multiple access with collision avoidance	CSMA/CD	... with collision detection
DCF	distributed coordination function	PCF	point coordination function
RTS/CTS	request to send / clear to send	CCK	complementary code keying
WEP, WPA	wired equivalent privacy, WiFi protected access	U-NII	unlicensed national information infrastructure band
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.

Short	Means	Short	Means
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
PCN	personal communication network	SMS	short message service
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
BSS	base station subsystem	NSS	network and switching subsystem
OSS	operation support subsystem	OMC	operation and maintenance centre
HLR	home location register	VLR	visitor location register
AuC	authentication centre	EIR	equipment identity register
CEIR	central EIR	ME	mobile equipment
SIM	subscriber identity module	IMSI	international mobile subscriber identity
Short	Means	Short	Means
IMEI	international mobile equipment identity	TMSI	temporary mobile subscriber identity
Um, Abis, A	the three GSM interfaces	SS7	signalling system no. 7
ARFCN	absolute radio frequency channel number	TN, TS	time slot number, time slot
TCH	traffic channel	TCH/FS, TCH/HS	full rate, half rate speech TCH
CCH	control channel	CCCH	common control channel
DCCH	dedicated control channel	SDCCH	stand-alone dedicated control channel
SACCH	slow associated control channel	FACCH	fast associated control channel
FCCH	frequency correction channel	SCH	synchronisation channel
PCH	paging channel	RACH	random access channel
AGCH	access grant channel	TB, GP, SF	tail bits, guard period, stealing flag
VAD	voice activity detector	DTX	discontinuous transmission
DCS 1800	digital cellular system, the 1800 MHz GSM	CRC	cyclic redundancy check
ESN	electronic serial number	PCB	power control bit
Mcps, Ksps	mega-chips, kilo-symbols per second	PRAT	paging channel rate field
EVRC	enhanced variable rate coder	PDSN	packet data serving node
PCF (CDMA)	packet control function	RNC	radio network controller
Short	Means	Short	Means
EPS, EPC	evolved packet system, core	eNodeB	LTE base station
MME	mobility management entity	S-GW, P-GW	serving, packet data network gateway
HSS	home subscriber server	PCRF	policy and charging rules function
IMS, VoLTE	IP multimedia subsystem, voice over LTE	CoMP	coordinated multipoint
ITU-R	ITU radiocommunication sector	WRC	world radiocommunication conference
NTA	Nepal Telecommunications Authority	UGS, rtPS, nrtPS	WiMAX QoS service classes

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet

data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

Three clashes to watch: **FCC** is the *forward control channel* throughout these notes; the US regulator is always written out as the Federal Communications Commission, and **FCCH** is GSM's frequency correction channel, a different thing again. **BCH** is the Bose–Chaudhuri–Hocquenghem block code in **Ch 6** and GSM's *broadcast channel* in **Ch 8**. **SCH** is GSM's synchronisation channel, never a sub-channel.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 23 papers, 2070–2082 BS: **TOP /23** = topic asked in 6 or more papers · **HOT /23** = 3 to 5 papers · **PIN /23** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.